Exercise 1.2. *Let x and y be elements and assume that $x \neq y$. Prove that $(x, y) \neq (y, x)$.*

Exercise 1.3. *Let n be a positive integer (natural number). Define ordered n-tuples.*

Exercise 1.6. *Let S and T be sets. How can we define a relation between S and T mathematically?*

Exercise 1.12. *Show that $\sup E$ is unique.*

The **greatest lower bound**, or **infimum** of a set E which is bounded below is defined in the same manner

Example 1.15. (*a*) *Let A be the set of all positive rationals p such that $p^2 < 2$ and B be the set of all positive rationals such that $p^2 > 2$. That is,*

$$A = \{p \in \mathbb{Q} : p > 0 \quad \textit{and} \quad p^2 < 2\}$$

and

$$B = \{p \in \mathbb{Q} : p > 0 \quad \textit{and} \quad p^2 > 2\}.$$

Since A has upper bounds, A is bounded above. In fact, the upper bounds of A are exactly the members of B. Since B contains no smallest member, A has no least upper bound in $\mathbb{Q}$. Similarly, B is bounded below. The set of all lower bounds of B in $\mathbb{Q}$ consists of A and of all $r \in \mathbb{Q}$ with $r \leq 0$. Since A has no largest member, B has no greatest lower bound in $\mathbb{Q}$.

(*b*) *If $\alpha = \sup E$ exists, then α may or may not be a member of E. Let*

$$E_1 = \{r \in \mathbb{Q} : r < 0\}$$

and

$$E_2 = \{r \in \mathbb{Q} : r \leq 0\}.$$

Then

$$\sup E_1 = \sup E_2 = 0$$

and $0 \notin E_1$, $0 \in E_2$.

Exercise 1.17. *Define the* **greatest-lower-bound property**.

Exercise 1.19. *Prove that the greatest-lower-bound property implies the least-upper-bound property.*

Exercise 1.27 (Difficult?). *Let $\mathbb{R}_1$ and $\mathbb{R}_2$ be ordered fields which have the least-upper-bound property and contain $\mathbb{Q}$. Show that we can identify $\mathbb{R}_1$ and $\mathbb{R}_2$ in a certain sense. In other words, show that $\mathbb{R}$ is the unique ordered field which has the least-upper-bound property and contains $\mathbb{Q}$ in a certain sense.*

Exercise 1.30. *Is it natural to define* $\inf \emptyset = \infty$*?*

Exercise 1.31. *Define*

$$\inf \emptyset = \infty \quad \textit{and} \quad \sup \emptyset = -\infty.$$

Show that every subset of $\overline{\mathbb{R}}$ *has a greatest lower bound.*

Exercise 1.34 (Difficult?)**.** (*i*) *Can we give an order on* $\mathbb{R}^2$*?*
(*ii*) *Can we make* $\mathbb{R}^2$ *an ordered field?*